

LECTURE 12 (OCTOBER 28TH)

Remark. We continue with the final step of the Riemann mapping theorem. Let Σ be the set of all 1-1 analytic functions from D into U . We have proved if that $\phi \in \Sigma$ satisfies $\phi(D) \neq U$, then for every $z_0 \in D$ there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\phi'(z_0)|$.

(Step 3) If $z_0 \in D$ and $\eta_0 = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$, then there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$, which implies that $h(D) = U$.

Proof. We suggest that $\mathcal{A} = \{|\phi'(z_0)| \mid \phi \in \Sigma\}$ is bounded. Suppose that it is not bounded. Then for every $n \in \mathbf{N}$, there is $\phi_n \in \Sigma$ such that $|\phi'_n(z_0)| > n$. However, notice that $|\phi_n(z)| < 1$ for all $z \in D$, $n \in \mathbf{N}$ and is uniformly bounded. By Montel's theorem, there is a subsequence $\{\phi_{n_k}\}$ which converges uniformly on every compact subset of D . Let $h(z) = \lim_{k \rightarrow \infty} \phi_{n_k}(z)$ for $z \in D$. Then, $h(z)$ is analytic in D and $\lim_{n \rightarrow \infty} \phi'_{n_k}(z) = h'(z)$ for every $z \in D$ so that $h'(z_0) = \lim_{k \rightarrow \infty} \phi'_{n_k}(z_0)$ which means that

$$|h'(z_0)| = \lim_{k \rightarrow \infty} |\phi'_{n_k}(z_0)| \geq \lim_{k \rightarrow \infty} n_k = \infty$$

which is a contradiction.

Knowing that $\eta_0 > 0$ (since $\phi' \neq 0$ for every $\phi \in \Sigma$) there is a subsequence $\{\phi_n\}$ in Σ such that $\lim_{n \rightarrow \infty} |\phi'_n(z_0)| = \eta_0$. Since $|\phi_n(z)| \leq 1$ for all $z \in D$, there is a subsequence $\{\phi_{n_k}\}$ which converges uniformly on all compact subsets of D to an analytic function h on D . Since $\lim_{k \rightarrow \infty} \phi'_{n_k}(z) = h'(z)$ for every $z \in D$, we have

$$|h(z_0)| = \lim_{k \rightarrow \infty} |\phi'_{n_k}(z_0)| = \eta_0$$

It remains to prove that $h \in \Sigma$. For every $z \in D$, $|h(z)| = \lim_{k \rightarrow \infty} |\phi_{n_k}(z)| \leq 1$ for all $z \in D$ as ϕ_{n_k} is a function from D to U . If $|h(z_1)| = 1$ for some $z_1 \in D$, by the maximum modulus theorem or the open mapping theorem, h is constant but $|h'(z_0)| = \eta_0 > 0$. This is again a contradiction. Thus, $h : D \rightarrow U$. Since ϕ_n is 1-1 analytic and $\phi_n \rightarrow h$ uniformly on all compact sets, we get that h is 1-1. $\square$

Example. Assume that $h(z_0) = 0$. If $h(z_0) = \beta \neq 0$ then $f = \phi_\beta \circ h \in \Sigma$ satisfies

$$|f'(z_0)| = |\phi'_\beta(\beta)h'(z_0)| = \frac{h'(z_0)}{1 - |\beta|^2} > |h'(z_0)|$$

Example. Prove or disprove whether for a $f : D \rightarrow U$ is analytic when $|f'(z_0)| \leq |h'(z_0)|$.

Proof. This is true. Define $g = f \circ h^{-1}$. Then, $g : U \rightarrow U$ is analytic.

$$g'(0) = f'(z_0)(h^{-1})'(0) = f'(z_0) \frac{1}{h'(z_0)} = \frac{f'(z_0)}{h'(z_0)}$$

since $|g'(0)| \leq 1$ which implies that $|f'(z_0)| \leq |h'(z_0)|$. □

Theorem. If $g : U \rightarrow U$ is 1-1 onto analytic with $g(0) = 0$ and $g'(0) > 0$, then $g(z) = z$.

Theorem. If D is simply connected ($D \neq \mathbf{C}$) then for every $z_0 \in D$, there is a unique 1-1 analytic function h from D onto U satisfying $h(z_0) = 0$ and $h'(z_0) > 0$.

Proof. If $h \in \Sigma$ satisfies $|h'(z_0)| = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$, then $h(D) = U$ and $h(z_0) = 0$. If there exists f and g both satisfying the conditions above, define $h = f \circ g^{-1}$ and $h(0) = 0$ and $h'(0) = h'(z_0)/g'(z_0) > 0$ giving $h(z) = z$. □

LECTURE 13 (OCTOBER 30TH)

Theorem. Let $h'(z_0) = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$ then $h(\Omega) = U$ and $h \in \Sigma$. In addition, $h(z_0) = 0$. In addition, if we add the condition that $h'(z_0) > 0$, such h is uniquely determined.

Proof. If $h'(z_0) = re^{i\theta}$, then $f(z) = e^{-i\theta}h(z)$ satisfies $f \in \Sigma$, $f(\Omega) = U$, $f(z_0) = 0$ and $f'(z_0) = r > 0$. Let's prove uniqueness. If there exists f and g satisfying all conditions, then define

$$h(z) = f(g^{-1}(z))$$

and $h : U \rightarrow U$ is 1-1 analytic and onto.

$$h'(0) = \frac{f'(z_0)}{g'(z_0)} > 0$$

and $h(0) = 0$. This implies that $h(z) = 0$ and that $f = g$. □

Theorem. Let D be simply connected and $D \neq \mathbf{C}$. Then there is $\phi : D \leftarrow U$ that is 1-1 analytic and onto. Recall that if ϕ is 1-1, $\phi' \neq 0$ and is conformal. This tells us that there exists $\psi = \phi^{-1} : U \rightarrow D$ that is still 1-1 analytic and onto, thus conformal. Define $A(D)$ be the set of all analytic functions on D . Define $\Phi : A(D) \rightarrow A(U)$ by $\Phi(f) = f \circ \psi$. Then, Φ becomes a ring isomorphism.

Remark. Consider $\mathbf{C}^2 = \mathbf{C} \times \mathbf{C}$. The simplest domains in $\mathbf{C}^2$ are B_2 and U^2 .

$$B_2 = \{z = (z_1, z_2) \in \mathbf{C}^2 \mid \|z\| < 1\} \quad \& \quad U^2 = U \times U$$

each called a bidisc and a unit ball. The norm $\|z\| = \langle z, z \rangle^{1/2}$ is defined as

$$\langle z, z \rangle^{1/2} = \langle (z_1, z_2), (z_1, z_2) \rangle^{1/2} = \left(z_1 \bar{z}_1 + z_2 \bar{z}_2 \right)^{1/2} = \left(|z_1|^2 + |z_2|^2 \right)^{1/2}$$

Theorem. (Residue theorem) If f is analytic inside and on a POSCC C except for isolated singularities at $z_1, \dots, z_n$ inside C , then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}_{z=z_k} f(z)$$

If $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ in $0 < |z - z_k| < \delta$ (Laurent series of f at z_k), then

$$a_{-1} = \text{Res}_{z=z_k} f(z)$$

If $\lim_{z \rightarrow z_0} (z - z_0)^k f(z) = A \neq 0$, we say that f has a pole of order k at z_0 . If f has a simple pole (order 1) at z_0 , then $\text{Res}_{z=z_0} f(z) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$. If

$$f(z) = \frac{p(z)}{q(z)}$$

(p, q analytic at z_0) has a simple pole at z_0 ($p(z_0) \neq 0$, $q(z_0) = 0$, and $q'(z_0) \neq 0$), then

$$\text{Res}_{z=z_0} f(z) = \lim_{z \rightarrow z_0} (z - z_0) \frac{p(z)}{q(z)} = \lim_{z \rightarrow z_0} \frac{p(z)}{\frac{q(z) - q(z_0)}{z - z_0}} = \frac{p(z_0)}{q'(z_0)}$$

Example. Evaluate for $0 < \alpha < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1 + e^x} dx$$

Proof. We need to take a rectangular contour C_R from $-R$ to R and height $2\pi i$ notice that the above function has a residue at πi . Take

$$f(z) = \frac{\alpha z}{1 + e^z}$$

For every $R > 0$,

$$\int_{C_R} f(z) dz = 2\pi i \text{Res}_{z=\pi i} f(z) = 2\pi i \frac{e^{\alpha \pi i}}{e^{\pi i}} = -2\pi i e^{\alpha \pi i}$$

On the other hand,

$$\int_{C_R} f(z) dz = \int_{-R}^R \frac{e^{\alpha x}}{1+e^x} dx + \int_0^{2\pi} \frac{e^{\alpha(R+iy)}}{1+e^{R+iy}} i dy + \int_R^{-R} \frac{e^{\alpha(x+2\pi i)}}{1+e^{x+2\pi i}} dx + \int_{2\pi}^0 \frac{e^{\alpha(-R+iy)}}{1+e^{-R+iy}} i dy$$

The second term goes to zero as $R \rightarrow \infty$ and

$$-2\pi i e^{\alpha\pi i} = \int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x} dx [1 - e^{2\alpha\pi i}]$$

resulting in

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x} dx = -\frac{2\pi i}{1 - e^{2\alpha\pi i}} = \frac{\pi}{\frac{e^{\alpha\pi i} - e^{-\alpha\pi i}}{2i}} = \frac{\pi}{\sin \alpha\pi}$$

If we further put $t = e^x$, $dx = e^{-x}$, and $dt/t = dx$. We therefore have

$$\int_0^{\infty} \frac{t^{\alpha-1}}{1+t} dt = \frac{\pi}{\sin \alpha\pi}$$

□

LECTURE 14 (NOVEMBER 4TH)

Definition. (Fourier transform of f on $\mathbf{R}$) The Fourier transform of f on $\mathbf{R}$ is defined as

$$\hat{f}(t) = \mathcal{F}(f)(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx$$

then

$$f(x) \sim \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{itx} dt$$

for a very huge class of functions f , the above is an equality. If we define

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-2\pi ixt} dx$$

then

$$f(x) \sim \int_{-\infty}^{\infty} \hat{f}(t) e^{2\pi ixt} dt$$

(this still works for f on $\mathbf{R}^n$) We also conventionally use

$$\hat{f}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-itx} dx$$

then

$$f(x) \sim \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt$$

Lastly, we also use

$$\hat{f}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-itx} dx$$

then

$$f(x) \sim \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt$$

The above works for $f \in L^1$, meaning that $\int_{-\infty}^{\infty} |f(x)| dx < \infty$. We can also define the above as

$$\hat{f}(t) = \lim_{A \rightarrow \infty} \int_{-A}^A f(x) e^{-itx} dx$$

when the limits exists.

Example. (Example of transform not being L^1) Consider the L^1 function

$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & |x| > 1 \end{cases}$$

the Fourier transform is then

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx = \int_{-1}^1 e^{-itx} dx = -\frac{1}{it} e^{-itx} \Big|_{-1}^1 = -\frac{1}{it} [e^{-it} - e^{it}] = \frac{2 \sin t}{t}$$

(which is 2 at $t = 0$) Notice that

$$\int_0^{N\pi} \left| \frac{\sin t}{t} \right| dt = \sum_{k=0}^{N-1} \int_{k\pi}^{(k+1)\pi} \left| \frac{\sin t}{t} \right| dt$$

where

$$\int_{k\pi}^{(k+1)\pi} \left| \frac{\sin t}{t} \right| dt \geq \int_{k\pi + \frac{\pi}{6}}^{k\pi + \frac{5\pi}{6}} \left| \frac{\sin t}{t} \right| dt \geq \frac{1}{2(k+1)\pi} \frac{2\pi}{3} = \frac{1}{3(k+1)}$$

and that the result is not L^1 . Is the inverse transform of $g(t) = 2 \sin / t$

$$\lim_{A \rightarrow \infty} \int_{-A}^A g(t) e^{ixt} dt = f(x)$$

is a good question.

Example. Let $a > 0$ and consider the L^1 function

$$f(x) = \frac{1}{x^2 + a^2}$$

and

$$\hat{f}(t) = \int_{-\infty}^{\infty} \frac{e^{-itx}}{x^2 + a^2} dx$$

Proof. Define

$$f(z) = \frac{e^{-itz}}{z^2 + a^2}$$

and use the residue theorem. In the upper hemisphere contour C_R with $R > a$,

$$\int_{C_R} f(z) dz = 2\pi i \operatorname{Res}_{z=ai} f(z) = 2\pi i \frac{e^{-it(ai)}}{2ai} = \frac{\pi}{a} e^{at}$$

Recall that for a simple pole,

$$\operatorname{Res}_{z=z_0} \left(\frac{p(z)}{q(z)} \right) = \frac{p(z_0)}{q'(z_0)}$$

The above is equal to

$$\int_{-R}^R \frac{e^{-itx}}{x^2 + a^2} dx + \int_0^\pi \frac{e^{-itRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} iRe^{i\theta} d\theta$$

The absolute value of the numerator is

$$\left| e^{-itRe^{i\theta}} \right| = \left| e^{-it(R \cos \theta + iR \sin \theta)} \right| = e^{tR \sin \theta}$$

and the second term of the above vanishes at $R \rightarrow \infty$ if the above is bounded and $t \leq 0$.

In the case that $t > 0$, we take the lower hemisphere and the pole is at $-ai$.

$$\int_{C_R} f(z) dz = -2\pi i \operatorname{Res}_{z=-ai} f(z) = 2\pi i \frac{e^{-it(-ai)}}{-2ai} = \frac{\pi}{a} e^{-at}$$

Likewise,

$$\int_{C_R} f(z) dz = \int_{-R}^R \frac{e^{-itx}}{x^2 + a^2} dx + \int_0^{-\pi} \frac{e^{-itRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} iRe^{i\theta} d\theta$$

and

$$\left| e^{-itRe^{i\theta}} \right| = e^{tR \sin \theta} < 1$$

since $-\pi\theta < \theta < 0$. Thus,

$$\hat{f}(t) = \int_{-\infty}^{\infty} \frac{e^{-itx}}{x^2 + a^2} dx = \frac{\pi}{a} e^{-a|t|}$$

which is L^1 . Let $a = 1$ and try to invert the above. If $g(t) = \pi e^{-|t|}$ then

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\infty}^{\infty} g(t) e^{ixt} dt &= \frac{1}{2} \int_{-\infty}^{\infty} e^{-|t|} e^{ixt} dt = \frac{1}{2} \left[\int_{-\infty}^0 e^t e^{ixt} dt + \int_0^{\infty} e^{-t} e^{ixt} dt \right] \\
&= \frac{1}{2} \left[\int_{-\infty}^0 e^{(1+ix)t} dt + \int_0^{\infty} e^{(-1+ix)t} dt \right] \\
&= \frac{1}{2} \left[\frac{e^{(1+ix)t}}{1+ix} \right]_{t=-\infty}^0 + \frac{1}{2} \left[\frac{e^{(-1+ix)t}}{-1+ix} \right]_{t=0}^{\infty} \\
&= \frac{1}{2} \left[\frac{1}{1+ix} \right] + \frac{1}{2} \left[\frac{1}{1-ix} \right] \\
&= \frac{1}{2} \left(\frac{1}{1+ix} + \frac{1}{1-ix} \right) = \frac{1}{1+x^2}
\end{aligned}$$

□

Example. Let $f(x) = e^{-x^2/2}$. Then,

$$\begin{aligned}
\hat{f}(t) &= \int_{-\infty}^{\infty} f(x) e^{-itx} dx = \int_{-\infty}^{\infty} e^{-x^2/2} e^{-itx} dx \\
&= \int_{-\infty}^{\infty} e^{-(x^2+2itx)} dx = \left[\int_{-\infty}^{\infty} e^{-(x+it)^2/2} dx \right] e^{-t^2/2}
\end{aligned}$$

which is $\sqrt{2\pi}$ when $t = 0$. When $t > 0$, we take the rectangular contour with base $2R$ and height $x + it$. By Cauchy,

$$0 = \int_{C_R} e^{-z^2/2} dz = \int_{-R}^R e^{-x^2/2} dx + (\text{II}) + (\text{III}) + (\text{IV})$$

We see that

$$(\text{III}) = \int_R^{-R} e^{-(x+it)^2/2} dt$$

and

$$(\text{II}) = \int_0^t e^{(R+iy)^2/2} i dy$$

We now use the fact that the absolute value of (II) and (III) goes to zero as $R \rightarrow \infty$. We finally have

$$\int_{-\infty}^{\infty} e^{-(x+it)^2/2} dx = \sqrt{2\pi}$$

with

$$\hat{f}(t) = \sqrt{2\pi} e^{-t^2/2}$$

LECTURE 15 (NOVEMBER 6TH)

Recall. We have previously shown the result (for $0 < \alpha < 1$)

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1 + e^x} dx = \frac{\pi}{\sin \alpha \pi}$$

where the change of variables $x = t^\alpha$ gives

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \alpha \pi}$$

Now put $x = t^\beta$ for $0 < \beta < \infty$.

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \int_0^\infty \frac{t^{\alpha\beta-1}}{1+t^\beta} \beta t^{\beta-1} dt = \beta \int_0^\infty \frac{t^{\alpha\beta-1}}{1+t^\beta} dt = \frac{\pi}{\alpha \pi}$$

Take $\alpha = 1/\beta$ such that $\beta > 1$. We get

$$\int_0^\infty \frac{1}{1+t^\beta} dt = \frac{1}{\beta \sin \frac{\pi}{\beta}}$$

for $\beta > 1$.

Recall. We have considered Fourier transform of

$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & |x| > 1 \end{cases}$$

which was 2 times the sinc function

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx = \frac{2 \sin t}{t}$$

We want to check that the inverse transform properly gives the original function.

Example. Find $\lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x} e^{itx} dx$. Notice that the kernel is an entire function, and that

$$\int_{-R}^R f(z) dz = \int_{C_R} f(z) dz = \int_{C_R} \frac{e^{iz} - e^{-iz}}{2iz} e^{itz} dz$$

where C_R is defined as contour that avoids the origin. Then,

$$\begin{aligned} \int_{-R}^R f(z) dz &= \int_{C_R} \frac{1}{2i} \left[\frac{e^{i(t+1)z} - e^{i(t-1)z}}{z} \right] dz \\ &= \frac{1}{2i} \left[\int_{C_R} \frac{e^{i(t+1)z}}{z} dz - \int_{C_R} \frac{e^{i(t-1)z}}{z} dz \right] \\ &= \varphi_R(t+1) - \varphi_R(t-1) \end{aligned}$$

where $\varphi_R(t) = \frac{1}{2i} \int_{C_R} \frac{e^{itz}}{z} dz$. Notice that

$$\left| \frac{e^{itz}}{z} \right| = \left| \frac{e^{itRe^{i\theta}}}{Re^{i\theta}} iRe^{i\theta} \right| = |e^{-tR \sin \theta}|$$

To evaluate

$$\varphi_R(t) = \frac{1}{2i} \int_{C_R} \frac{e^{itz}}{z} dz$$

where for $t < 0$, we choose a POSCC so that θ is from 0 to $-\pi$. We find

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{2i} \frac{e^{itz}}{z} dz = 0$$

For $t > 0$, we choose the POSCC $\Gamma_R = C_R \cup \{\text{upper semi circle}\}$. We find

$$\int_{\Gamma_R} \frac{e^{itz}}{z} dz = 2\pi i$$

thus

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{2i} \frac{e^{itz}}{z} dz = \pi$$

Our goal was

$$\begin{aligned} \lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x} e^{itx} dx &= \lim_{R \rightarrow \infty} \int_{C_R} \frac{\sin x}{x} e^{itx} dx = \lim_{R \rightarrow \infty} [\varphi_R(t+1) - \varphi_R(t-1)] \\ &= \lim_{R \rightarrow \infty} \varphi_R(t+1) - \lim_{R \rightarrow \infty} \varphi_R(t-1) \\ &= \begin{cases} \pi & -1 < t < 1 \\ \pi/2 & t = 0 \\ 0 & |t| > 1 \end{cases} \end{aligned}$$

Now, $\hat{f}(t) = 2 \sin t/t$ gives

$$\lim_{R \rightarrow \infty} \int_{-R}^R \hat{f}(t) e^{ixt} dt$$

We can further find that $\lim_{R \rightarrow \infty} \varphi(0) = (i\pi)/2i = \pi/2$.

Theorem. (Big theorem)

- (i) If f and f' are L^1 , $\hat{f}'(t) = it\hat{f}(t)$
- (ii) If $f(x)$ and $xf(x)$ are in L^1 , then

$$(xf(x))^\wedge(t) = i(\hat{f})'(t)$$

and

$$\mathcal{F}(xf(x))(t) = i(\hat{f})'(t)$$

Proof. (I)

$$\hat{f}'(t) = \int_{-\infty}^{\infty} f'(x)e^{-itx} dx = \left[f(x)e^{-itx} \right]_{x=-\infty}^{\infty} + \int_{-\infty}^{\infty} f(x)ite^{-itx} dx$$

However, the first term is zero as

$$\lim_{x \rightarrow \infty} f(x) = f(0) + \int_0^{\infty} f'(x) dx$$

where $\int_0^{\infty} f'(x) dx = \lim_{x \rightarrow \infty} f(x) - f(0)$ and if $\lim_{x \rightarrow \infty} f(x) = \alpha$, $\alpha = 0$. (II)

$$\frac{d}{dt} \hat{f}(t) = \frac{d}{dt} \int_{-\infty}^{\infty} f(x)e^{-ixt} dt = \int_{-\infty}^{\infty} -ixf(x)e^{-ixt} dt = -i \left[xf(x) \right]'(t)$$

□

Example. Last class we saw that for $f(x) = \exp(-x^2/2)$ implies $\hat{f}(t) = \sqrt{2\pi} \exp(-t^2/2)$. Then,

$$f'(x) = -xe^{-x^2/2}$$

so that

$$f'(x) = -xf(x)$$

Applying the Fourier transform,

$$it\hat{f}(t) = -i(\hat{f})'(t)$$

and

$$(\hat{f})'(t) = -t\hat{f}(t)$$

LECTURE 16 (NOVEMBER 11TH)

Theorem. $f \in L^1$ implies that $\int_{-\infty}^{\infty} |f(x)| dx < \infty$. Let $f, f' \in L^1$. Then,

$$\hat{f}'(t) = it\hat{f}(t)$$

and $f, xf(x) \in L^1$ implies that

$$\mathcal{F}(xf(x))(t) = i(\hat{f})'(t)$$

Example. If $f(x) = e^{-ax^2/2}$, $a > 0$, then $f'(x) = -axf(x)$. Take the Fourier transform,

$$it\hat{f}(t) = -ai(\hat{f})'(t)$$

so that $(\hat{f})'(t) = -t\hat{f}(t)/a$ denotes $g(t) = \hat{f}(t)$. Then $g'(t) = -tg(t)/a$ and

$$g(0) = \hat{f}(0) = \int_{-\infty}^{\infty} e^{-ax^2/2} dx = \sqrt{\frac{2\pi}{a}}$$

If I is an open interval containing 0 such that $g(t) > 0$ on I . Then for $t \in I$,

$$\frac{g'(t)}{g(t)} = -\frac{t}{a}$$

implies that

$$\ln g(t) = -\frac{t^2}{2a} + C$$

and $g(t) = C_1 e^{-t^2/2a}$. Put $t = 0$, and

$$g(t) = \sqrt{\frac{2\pi}{a}} e^{-t^2/2a}$$

on I . If $I \neq \mathbf{R}$, then there is an endpoint t_0 of I such that $g(t_0) \leq 0$. Put

$$g(t_0) = \lim_{t \rightarrow t_0} g(t) = \lim_{t \rightarrow t_0} \sqrt{\frac{2\pi}{a}} e^{-t^2/2a} = \sqrt{\frac{2\pi}{a}} e^{-t_0^2/2a} > 0$$

which is a contradiction.

$$\mathcal{F}(e^{-ax^2/2})(t) = \sqrt{\frac{2\pi}{a}} e^{-t^2/2a}$$

for $a > 0$.

Definition. (Convolution of f and g) Let

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) dy$$

wherever it is defined.

Theorem. If $f, g \in L^1$ and $h = f * g$, then $h \in L^1$.

Example. Let

$$f(x) = \begin{cases} 1/\sqrt{x} & (0, 1) \\ 0 & \text{elsewhere} \end{cases} \quad g(x) \in L^1$$

also,

$$f(x)g(x) = \begin{cases} 1/x & (0, 1) \\ 0 & \text{elsewhere} \end{cases}$$

which is not L^1 . Thus, L^1 isn't closed under multiplication.

Lemma.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dy \, dx$$

if

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)| \, dx \, dy < \infty$$

Also, if $f(x, y) \geq 0$ for all x, y , then the above also holds.

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} |h(x)| \, dx &= \int_{-\infty}^{\infty} (f * g(x)) \, dx = \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} f(x-y)g(y) \, dy \right| \, dx \\ &\leq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x-y)||g(y)| \, dy \, dx \leq \int_{-\infty}^{\infty} |g(y)| \, dy \int_{-\infty}^{\infty} |f(x-y)| \, dx < \infty \end{aligned}$$

□

Theorem. If $f, g \in L^1$, then $f * g = g * f$.

$$(f * g)(t) = \int_{-\infty}^{\infty} f(x-y)g(y) \, dy = \int_{-\infty}^{\infty} f(t)g(x-t)(-dt) = \int_{-\infty}^{\infty} g(x-t)f(t) \, dt = (g * f)(x)$$

by $x - y = t$ and $dy = -dt$. Thus,

$$(f * g)(x) = \int_{-\infty}^{\infty} g(x-y)f(y) \, dy$$

Remark. The meaning of $f * g$ is the weighted average of f using g . For $a > 0$, consider

$$g_a(x) = \begin{cases} 1/2a & -a < x < a \\ 0 & \text{elsewhere} \end{cases}$$

then,

$$g_a(x) \geq 0 \quad \& \quad \int_{-\infty}^{\infty} g_a(x) \, dx = 1 \quad \& \quad g_a(-x) = g_a(x)$$

then

$$(f * g_a)(x) = \int_{-\infty}^{\infty} g_a(x-y)f(y) \, dy$$

where

$$g_a(x-y) = \begin{cases} 1/2a & x-a < y < x+a \\ 0 & \text{elsewhere} \end{cases}$$

Notice how

$$(f * g)(x) = \frac{1}{2a} \int_{x-a}^{x+a} f(y) \, dy$$

taken to $a \rightarrow 0^+$ leaves us with $(f * g_a)(x) \rightarrow f(x)$ if $f(x)$ is continuous at x .

$$\lim_{a \rightarrow 0^+} (f * g_a)(x) = \frac{1}{2}(f(x_+) + f(x_-))$$

Remark. If $\phi \geq 0$,

$$\int_{-\infty}^{\infty} \phi(x) dx = 1$$

and $\phi(-x) = \phi(x)$, then

$$\phi_\varepsilon(x) = \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right)$$

with $\varepsilon > 0$ satisfies

$$\int_{-\infty}^{\infty} \phi_\varepsilon(x) dx = \int_{-\infty}^{\infty} \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right) dx = \int_{-\infty}^{\infty} \phi(t) dt = 1$$

and

$$\lim_{\varepsilon \rightarrow 0^+} (f * \phi_\varepsilon)(x) = \frac{1}{2}(f(x_+) + f(x_-))$$

For example, we have the Cauchy distribution

$$\phi(x) = \frac{1}{\pi} \frac{1}{1 + x^2}$$

and the normal distribution

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

which is the smoothing function. Notice that

$$(f * \phi)(x) = (\phi * f)(x) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x-y)^2/2} f(y) dy$$

(the exponential function is $C^\infty(\mathbf{R})$ and is nice)

$$(f * \phi_\varepsilon)(x) = \int_{-\infty}^{\infty} \frac{1}{\varepsilon \sqrt{2\pi}} e^{-|x-y|^2/2\varepsilon^2} f(y) dy$$

which implies

$$\frac{1}{2}|f(x_+) + f(x_-)|$$

and $\varepsilon \rightarrow 0$. The above is exactly $f(x)$ on every norm. Note that

$$\mathcal{F}\left(e^{-ax^2/2}\right)(t) = \sqrt{\frac{2\pi}{a}} e^{-t^2/2a}$$

and $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$. Then for $f \in L^1$,

$$(f * \phi_\varepsilon)(x) = \int_{-\infty}^{\infty} \frac{1}{\varepsilon \sqrt{2\pi}} e^{-|x-y|^2/2\varepsilon^2} f(y) dy$$

with

$$\phi_\varepsilon(x) = \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right) = \frac{1}{\varepsilon\sqrt{2\pi}} e^{-x^2/2\varepsilon^2}$$

Theorem. We want to verify the equality

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt = \frac{1}{2} (f(x_+) + f(x_-))$$

if $f \in L^1$ and is piecewise continuous.

Proof. Indeed,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{-ity} dy e^{ixt} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{i(x-y)t} dy dt$$

This is a dead lock ($|e^{i(x-y)t}| = 1$ and is not L^1 with respect to t). Instead,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{-\varepsilon^2 t^2/2} e^{ixt} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) e^{-ity} dy e^{-\varepsilon^2 t^2/2} e^{ixt} dt$$

and

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{-\varepsilon^2 t^2/2} e^{i(x-y)t} dy dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) \int_{-\infty}^{\infty} e^{-\varepsilon^2 t^2/2} e^{-i(y-x)t} dt dy$$

where the most right integral satisfies

$$\mathcal{F}\left(e^{-\varepsilon^2 t^2/2}\right)(y-x) = \frac{\sqrt{2\pi}}{\varepsilon} e^{-(y-x)^2/2\varepsilon^2}$$

resulting in

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) \frac{\sqrt{2\pi}}{\varepsilon} e^{-(y-x)^2/2\varepsilon^2} dy = (f * \phi_\varepsilon)(x) \rightarrow \frac{1}{2} (f(x_+) + f(x_-))$$

as $\varepsilon \rightarrow 0$ and

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{-\varepsilon^2 t^2/2} e^{itx} dt \rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{itx} dt$$

as $\varepsilon \rightarrow 0$. □

LECTURE 17 (NOVEMBER 13TH)

Recall. Consider $f \in L^1$ be piecewise continuous, and define

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

We define

$$\phi_\epsilon(x) = \frac{1}{\epsilon} \phi_\epsilon\left(\frac{x}{\epsilon}\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\epsilon^2}}$$

this implies that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{itx} dt = \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\hat{f}(t) e^{-\epsilon^2 t^2 / 2} \right) e^{ixt} dx = \lim_{\epsilon \rightarrow 0} (f * \phi_\epsilon)(x) = \frac{1}{2} (f(x+) + f(x-))$$

Note that $\phi(x)$ is in the Schwartz class, satisfying

$$\sup_{x \in \mathbf{R}} \left(x^k \phi^{(n)}(x) \right) < \infty$$

for all $k, n \geq 0$.

Definition. (Schwartz class $\mathcal{S}$) A function is in the Schwartz class $f \in \mathcal{S}$ if and only if $f \in C^\infty(\mathbf{R})$ and

$$\sup_{x \in \mathbf{R}} |x^k f^{(n)}(x)| < \infty$$

If $f \in \mathcal{S}$ then $(\hat{f})^\vee = f$ where

$$g^\vee(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{ixt} dt$$

Thus, $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ defined by $\mathcal{F}(f) = \hat{f}$ is an isomorphism.

Recall. Recall that

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) dy = \int_{-\infty}^{\infty} f(y)g(x-y) dy = (g * f)(x)$$

if $f, g \in L^1$ then $f * g \in L^1$.

Theorem. If $f, g \in L^1$, then $(f * g)^\wedge(t) = \hat{f}(t)\hat{g}(t)$.

Proof.

$$\begin{aligned} (f * g)^\wedge(t) &= \int_{-\infty}^{\infty} (f * g)(x) e^{-ixt} dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-y)g(y) dy e^{-ixt} dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-y)g(y) e^{-ixt} dy dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-y) e^{-i(x-y)t} g(y) e^{-iyt} dx dy \\ &= \int_{-\infty}^{\infty} g(y) e^{-iyt} dy \int_{-\infty}^{\infty} f(x-y) e^{-i(x-y)t} dx \\ &= \hat{g}(t) \hat{f}(t) \end{aligned}$$

□

Theorem. (Heat equation) Let's assume $u(x, t) \in \mathcal{S}$ for every fixed t .

$$\begin{cases} u_{xx} = u_t & t > 0 \\ u(x, 0) = f(x) \end{cases}$$

Define

$$g(w, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ixw} dx$$

which is the Fourier transform of $u(x, t)$. Recall that $\hat{f}'(t) = it\hat{f}(t)$.

$$\int_{-\infty}^{\infty} u_t(x, t) e^{-ixw} dx = \frac{\partial g}{\partial t}(w, t)$$

Together this implies that in the Fourier space,

$$-w^2 g = \frac{\partial g}{\partial t}$$

and the initial condition becomes

$$g(w, 0) = \hat{f}(w)$$

Once fixing w , we see that the above is an ordinary differential equation with variable t .

$$g(w, t) = \hat{f}(w) e^{-iw^2 t}$$

Last class, we've also learned

$$\mathcal{F}(e^{-ax^2/2})(w) = \sqrt{\frac{2\pi}{a}} e^{-w^2/2a}$$

Now, for fixed $t > 0$, put $a = 1/2t$,

$$\mathcal{F}\left(\sqrt{\frac{1}{4\pi t}} e^{-x^2/4t}\right) = e^{-w^2 t}$$

Now define $h_t(x) = \sqrt{1/4\pi t} \exp(-x^2/4t)$, then $\hat{h}_t(w) = \exp(-w^2 t)$.

$$g(w, t) = \hat{u}(w, t) = \hat{f}(w) \hat{h}_t(w)$$

By the inversion theorem, together with

$$(f * h_t)^\wedge = \hat{f} \hat{h}_t$$

we have $u(x, t) = (f * h_t)(x)$

$$u(x, t) = \int_{-\infty}^{\infty} f(y) \frac{1}{\sqrt{2\pi t}} e^{-(x-y)^2/4t} dy$$

Recall. Let

$$g(t) = \begin{cases} e^{-1/t} & t > 0 \\ 0 & t \leq 0 \end{cases}$$

then $g \in C^\infty(\mathbf{R})$. Also, let

$$h(t) = \begin{cases} e^{-1/(1-t^2)} & -1 < t < 1 \\ 0 & |t| \geq 1 \end{cases}$$

then $h \in C_c^\infty(\mathbf{R})$, meaning that C^∞ with a compact support. $f \in C_c(\mathbf{R})$ with support K when $f(x) = 0$ if $x \in K^c$. Finally, define $\psi : \mathbf{R} \rightarrow \mathbf{R}$ by

$$\psi(x) = \frac{h(x)}{\int_{-1}^1 h(t) dt}$$

then $\psi \in C_c^\infty(\mathbf{R})$ with support $[-1, 1]$ and $\int_{-\infty}^{\infty} \psi(x) dx = 1$. $\psi \in C_c^\infty(\mathbf{R})$ satisfies $\psi \geq 0$, $\psi(-x) = \psi(x)$, $\int_{-\infty}^{\infty} \psi(x) dx = 1$. Define

$$\psi_\varepsilon(x) = \frac{1}{\varepsilon} \psi\left(\frac{x}{\varepsilon}\right)$$

and $f \in L^1(\mathbf{R})$ and $f * \psi_\varepsilon \in C^\infty(\mathbf{R})$ with $f * \psi_\varepsilon \rightarrow f$.

Theorem. If f and g have a compact support, then $f * g$ also has a compact support.

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) dy$$

has a compact support.

Corollary. If $f \in L^1$ with a compact support, then $f * \psi_\varepsilon \in C_c^\infty(\mathbf{R}) \subset \mathcal{S}$.

Recall. If f satisfies $\int_{\mathbf{R}} |f|^p dx < \infty$ for $1 \leq p < \infty$, then for every $\varepsilon > 0$ there is $h \in C_c(\mathbf{R})$ such that $\int_{\mathbf{R}} |f - h|^p dx < \varepsilon$. This tells us that $C_c(\mathbf{R})$ is dense in $L^p(\mathbf{R})$ for $1 \leq p < \infty$.

Example. Let f satisfy $\int_{-\infty}^{\infty} |f(y)|^2 dy < \infty$. Define $g(x) = \overline{f(-x)}$. For $f \in \mathcal{S}$,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) dt = \left(\hat{f}\right)^\vee(0) = f(0)$$

and $\int_{-\infty}^{\infty} \hat{f}(t) dt = \sqrt{2\pi} f(0)$. Then,

$$\begin{aligned} (f * g)(0) &= \int_{-\infty}^{\infty} g(0-y)f(y) dy \\ &= \int_{-\infty}^{\infty} \overline{f(y)} f(y) dy = \int_{-\infty}^{\infty} |f(y)|^2 dy \end{aligned}$$

Thus,

$$\begin{aligned} \int_{-\infty}^{\infty} |f(y)|^2 dy &= (f * g)(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (f * g)^{\wedge}(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) \hat{g}(t) dt \\ \hat{g}(t) &= \int_{-\infty}^{\infty} g(x) e^{-ixt} dt = \int_{-\infty}^{\infty} \overline{f(-x)} e^{-ixt} dx = \int_{-\infty}^{\infty} \overline{f(y)} e^{iyt} (-dy) \\ &= \overline{\int_{-\infty}^{\infty} f(y) e^{-iyt} dy} = \overline{\hat{f}(t)} \end{aligned}$$

by inversion theorem if $f, g \in \mathcal{S}$. In conclusion,

$$\int_{-\infty}^{\infty} |f(y)|^2 dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(t)|^2 dt$$

holds for every f with $\int_{-\infty}^{\infty} |f(y)|^2 < \infty$. Plancherel theorem.

LECTURE 18

Recall. We have previously learned the Schwartz class $\mathcal{S}$ and that it is $\mathcal{S}$ is dense in L^1 . This implied that for $f \in L^1$, there is $\{f_n\}$ in $\mathcal{S}$ such that

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} |f_n - f| dx = 0$$

Also, if $f \in \mathcal{S}$, $\hat{f} \in \mathcal{S}$.

Theorem. (Riemann Lebesgue lemma) If $f \in L^1$, then

$$\lim_{|t| \rightarrow \infty} \hat{f}(t) = 0$$

Proof. Take any $\varepsilon > 0$, we have to find $M > 0$ satisfying if $|t| > M$, then $|\hat{f}(t)| < \varepsilon$. We use the fact that $\mathcal{S}$ is dense in L^1 and that $\hat{f}(t) = \hat{f}_n(t) + (\hat{f}(t) - \hat{f}_n(t))$ where $f_n \in \mathcal{S}$. For $f \in L^1$, there is $f_N \in \mathcal{S}$ such that

$$\int_{-\infty}^{\infty} |f - f_N| dx < \frac{\varepsilon}{2}$$

Since $\hat{f}_N \in \mathcal{S}$, there is $M > 0$ such that if $|t| > M$ then $|\hat{f}_N(t)| < \varepsilon/2$. And

$$\hat{f}_N(t) - \hat{f}(t) = \int_{-\infty}^{\infty} (f_N(x) - f(x)) e^{-itx} dx$$

so that

$$|\hat{f}_N(t) - \hat{f}(t)| \leq \int_{-\infty}^{\infty} |f_N(x) - f(x)| dx < \frac{\varepsilon}{2}$$

Hence, if $|t| > M$ then

$$|\hat{f}(t)| \leq |\hat{f}_N(t)| + |\hat{f}(t) - \hat{f}_N(t)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□

Corollary. If $f \in L^1$, then $\hat{f}$ is continuous on $\mathbf{R}$. Take $\{f_n\}$ in $\mathcal{S}$ such that

$$\int_{-\infty}^{\infty} |f_n - f| dx \rightarrow 0$$

then for every $t \in \mathbf{R}$

$$|\hat{f}_n(t) - \hat{f}(t)| \leq \int_{-\infty}^{\infty} |f_n(x) - f(x)| dx$$

That is, $\hat{f}_n \rightarrow \hat{f}$ uniformly on $\mathbf{R}$.

Definition. (Laplace transform of f) Let $f(t) = 0$ if $t < 0$, and continuous for $f \geq 0$. Lastly, let there exist $M, k, t_0 > 0$ such that

$$|f(t)| \leq M e^{kt}$$

for every $t > t_0$. For such f , the Laplace transform F of f is defined as

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

where $\operatorname{Re}(s) > k$ and $s \in \mathbf{C}$, analytic on the half plane $\operatorname{Re}(s) > k$. Indeed, if $s = \sigma + iw$ (where $\sigma > k$) then

$$F(s) = \int_0^{\infty} f(t) e^{-\sigma t} e^{-iwt} dt$$

which is $\hat{g}(w)$ where $g(t) = f(t) e^{-\sigma t}$ and is L^1 . We now apply the inversion theorem on

$$\hat{g}(w) = \int_0^{\infty} f(t) e^{-\sigma t} e^{-iwt} dt = \int_0^{\infty} f(x) e^{-\sigma x} e^{-iwx} dx$$

Then,

$$f(t) e^{-\sigma t} = g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{g}(w) e^{itw} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) e^{-\sigma x} e^{-iwx} dx e^{itw} dw$$

Hence,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_0^{\infty} f(x) e^{-(\sigma+iw)x} dx e^{(\sigma+iw)t} dw$$

Putting pack $s = \sigma + iw$ ($dw = ds/i$),

$$f(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \int_0^{\infty} f(x) e^{-sx} dx e^{st} ds = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} F(s) e^{st} ds$$

for $t > 0$.

Example. If $f(t) = 1 - e^{-t}$ for $t \geq 0$, zero elsewhere. Then

$$F(s) = \int_0^{\infty} (1 - e^{-t}) e^{-st} dt = \int_0^{\infty} e^{-st} dt - \int_0^{\infty} e^{-(s+1)t} dt = \frac{1}{s} - \frac{1}{s+1} = \frac{1}{s(s+1)}$$

Let's try the inversion theorem!

$$f(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s(s+1)} ds$$

and we the left hemisphere contour $C_R \cup \{x = \sigma\}$. For the right edge, we use $s = \sigma + iy$ ($-R \leq y \leq R$). For the round circumference, $s = \sigma + Re^{i\theta}$ with $\pi/2 \leq \theta \leq 3\pi/2$. Let $f(s) = e^{st}/s(1+s)$ Always,

$$\lim_{R \rightarrow \infty} \int_{C_R} f(s) ds = 0$$

must hold. In this particular case,

$$\int_{C_R} f(s) ds \rightarrow 0$$

as $R \rightarrow \infty$. Thus,

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s(1+s)} ds = 2\pi i \frac{1}{2\pi i} (1 - e^{-t}) = 1 - e^{-t}$$

Recall. Notice that

$$\int_0^{\pi} e^{-R \sin \theta} d\theta = 2 \int_0^{\pi/2} e^{-R \sin \theta} d\theta$$

and that $2\theta/\pi \leq \sin \theta \leq \theta$.

$$\int_0^{\pi/2} e^{-R \sin \theta} d\theta \leq \int_0^{\pi/2} e^{-R\theta/2\pi} d\theta = \frac{\pi}{2R} (1 - e^{-R}) \leq \frac{\pi}{2R}$$

and

$$\int_0^{\pi} e^{-R \sin \theta} d\theta \leq \frac{\pi}{R}$$

Example. Find

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s^2+1} ds = \operatorname{Res}_{z=i} f + \operatorname{Res}_{z=-i} f =$$

where $f(s) = e^{st}/(s^2+1)$. Notice that

$$\operatorname{Res}_{z=\pm i} f = \frac{p(\pm i)}{q'(\pm i)}$$

where $p(s) = e^{st}$ and $q(s) = s^2+1$. We thus have

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s^2+1} ds = \frac{e^{it}}{2i} - \frac{e^{-it}}{2i} = \sin t$$

Example. Find

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{2s^3 e^{st}}{s^4-4} ds$$

Note that $s^4-4=0$ implies $s = \pm\sqrt{2}, \pm i\sqrt{2}$.

$$\operatorname{Res}_{z=\alpha} f = \frac{2\alpha^3 e^{\alpha t}}{4\alpha^3} = \frac{e^{\alpha t}}{2}$$

and

$$\frac{e^{t\sqrt{2}}}{2} + \frac{e^{-t\sqrt{2}}}{2} + \frac{e^{-it\sqrt{2}}}{2} + \frac{e^{it\sqrt{2}}}{2} = \cosh \sqrt{2}t + \cos \sqrt{2}t$$

LECTURE 19 (NOVEMBER 20TH)

Remark. Consider the function

$$f(z) = \pi \cot \pi z = \frac{\pi \cos \pi z}{\sin \pi z}$$

which has a simple pole at each $n \in \mathbf{Z}$.

$$\operatorname{Res}_{z=n} f(z) = \frac{\pi \cos n\pi}{(\sin \pi z)'_{z=n}} = \frac{\pi \cos n\pi}{\pi \cos n\pi} = 1$$

Consider the square contour C_N and we find

$$\pi \cot \pi z = \frac{1}{z} \left(1 - \frac{\pi^2 z}{3} - \frac{\pi^4 z^4}{45} - \dots \right)$$

Example. We can use the above function to evaluate

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \xi(4)$$

Observe how

$$\int_{C_N} \frac{\pi \cot \pi z}{z^4} dz \rightarrow 0$$

as $N \rightarrow \infty$ due to parametrisation. This is in turn, due to the residue theorem,

$$2\pi i \sum_{k=-N}^N \operatorname{Res}_{z=k} \left[\frac{\pi \cot \pi z}{z^4} \right] = 2\pi i \left[2 \sum_{k=1}^N \frac{1}{k^4} + \operatorname{Res}_{z=0} \frac{\pi \cot \pi z}{z^4} \right]$$

and we expect the residue in the bracket to be

$$\operatorname{Res}_{z=0} \frac{\pi \cot \pi z}{z^4} = -\frac{\pi^4}{45}$$

and

$$\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$$

Example. For $a \notin \mathbf{Z}$, evaluate

$$\sum_{n=-\infty}^{\infty} \frac{1}{(z+a)^2}$$

Proof.

$$\int_{C_N} \frac{\pi \cot \pi z}{(n+a)^2} dz = 2\pi i \left[\sum_{k=-N}^N \operatorname{Res}_{z=k} \left(\frac{\pi \cot \pi z}{(z+a)^2} \right) + \operatorname{Res}_{z=-a} \left(\cdot \right) \right]$$

If f has a pole of order 2 at z_0 , then

$$\operatorname{Res}_{z=z_0} f(z) = \frac{1}{(k-1)!} \lim_{z \rightarrow z_0} \left[(z-z_0)^k f(z) \right]^{(k-1)}$$

Thus

$$\operatorname{Res}_{z=-a} \frac{\pi \cot \pi z}{(z+a)^2} = \lim_{z \rightarrow -a} [\pi \cot \pi z]' = [\pi(-\csc^2 \pi z)\pi]_{z=-a} = -\frac{\pi^2}{\sin^2 a\pi}$$

Thus,

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+a)^2} = \frac{\pi^2}{\sin^2 a\pi}$$

□

Example. Define

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + a^2}$$

for $a > 0$. Use

$$\int_{C_N} \frac{\pi \cot \pi z}{n^2 + a^2} dz \rightarrow 0$$

as $N \rightarrow \infty$. The above is equal to

$$2\pi i \left[\sum_{k=-N}^N \left(\cdot \right) + \operatorname{Res}_{z=ai} \left(\cdot \right) + \operatorname{Res}_{z=-ai} \left(\cdot \right) \right]$$

We have

$$\frac{1}{2} \left(\frac{\pi \cosh \pi a}{a \sinh \pi a} - \frac{1}{a^2} \right)$$

Theorem. For $z \notin \mathbf{Z}$,

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$$

For $w \in \mathbf{Z}$,

$$\int_{C_N} \frac{\pi \cot \pi z}{z(z-w)} dz = 2\pi i \left[\sum_{0 \neq k=-N}^N \operatorname{Res}_{z=k} \frac{\pi \cot \pi z}{z(z-w)} + \operatorname{Res}_{z=0} \frac{\pi \cot \pi z}{z(z-w)} + \operatorname{Res}_{z=w} \left(\cdot \right) \right]$$

which goes to 0 as $N \rightarrow \infty$. The first residue is equal to

$$\frac{1}{k(k-w)}$$

and the last is equal to

$$\frac{\pi \cot \pi w}{w}$$

For the second, we use partial fractions,

$$\frac{1}{z(z-w)} = -\frac{1}{w} \left(\frac{1}{z} - \frac{1}{z-w} \right)$$

and we have $-1/w^2$ for the second one. Gathering the results,

$$\int_{C_N} \frac{\pi \cot \pi z}{z(z-w)} dz = 2\pi i \left[\sum_{0 \neq k=-N}^N \frac{1}{k(k-w)} - \frac{1}{w^2} + \frac{\pi \cot \pi w}{w} \right]$$

Notice, however,

$$\sum_{k=1}^N \frac{1}{k(k-w)} + \sum_{k=-N}^{-1} \frac{1}{k(k-w)} = \sum_{k=1}^N \frac{1}{k} \left(\frac{1}{k-w} + \frac{1}{k+w} \right) = \sum_{k=1}^N \frac{1}{k} \frac{2k}{k^2 - w^2} = \sum_{k=1}^N \frac{2}{k^2 - w^2}$$

and

$$\pi \cot \pi w = \frac{1}{w} + \sum_{k=1}^{\infty} \frac{2w}{w^2 - k^2}$$

Example. (Reverse parametrisation)

$$\int_0^{2\pi} \frac{1}{1 + a \cos \theta} d\theta$$

with $0 < a < 1$. We do this through paramterising in the opposite direction, $C : |z| = 1$ and $z = e^{i\theta}$ ($0 \leq \theta \leq 2\pi$).

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2}$$

and $dz = ie^{i\theta} d\theta = iz d\theta$ implying $d\theta = dz/iz$. So,

$$\int_0^{2\pi} \frac{d\theta}{1 + a \cos \theta} = \int_{|z|=1} \frac{1}{1 + a \left(\frac{z + z^{-1}}{2} \right)} \frac{dz}{iz} = \frac{1}{ai} \int_{|z|=1} \frac{2 dz}{z^2 + \frac{2z}{a} + 1}$$

now, $z^2 + 2z/a + 1 = (z - z_1)(z - z_2)$ and

$$z_1 = -\frac{1}{a} + \sqrt{\frac{1}{a^2} - 1} = \frac{-1 + \sqrt{1 - a^2}}{a} \quad \& \quad z_2 = \frac{-1 - \sqrt{1 - a^2}}{a}$$

where the second is outside $|z| = 1$.

$$z_1 - z_2 = \frac{2\sqrt{1 - a^2}}{a}$$

and

$$2\pi i \frac{1}{ai} \operatorname{Res}_{z=z_1} \left(\cdot \right) = 2\pi i \frac{1}{ai} \frac{2}{z_1 - z_2} = \frac{2\pi}{\sqrt{1 - a^2}}$$

Example. Let's see another example of reverse parametrisation. For $n \in \mathbf{N}$, evaluate

$$\int_0^{2\pi} \sin^{2n} t dt = \int_{|z|=1} \left(\frac{z - z^{-1}}{2i} \right)^{2n} \frac{dz}{iz} = \frac{1}{i(2i)^{2n}} \int_{|z|=1} (z - z^{-1})^{2n} z^{-1} dz$$

Let $z = e^{it}$ for $0 \leq t \leq 2\pi$.

$$\sin t = \frac{z - z^{-1}}{2i}$$

with $dt = dz/iz$. We only need to find the coefficient of z^{-1} .

$$(z - z^{-1})^{2n} = \sum_{k=0}^{2n} {}^{2n}C_k z^k (-z^{-1})^{2n-k}$$

Finding the constant coefficient,

$$2\pi i \frac{1}{i} \frac{1}{(2i)^{2n}} {}^{2n}C_n (-1)^n = 2\pi \frac{(2n)!}{2^{2n} (n!)^2}$$

LECTURE 20 (NOVEMBER 25TH)

Remark. Let f be an entire function. The zero set of f is denoted as Z_f . If A is an infinite subset of $\mathbf{C}$ and $Z_f = A$ for some entire function f , then A must not have a limit point. If $A = \{a_n\} \subset \mathbf{C}$ with $\lim_{n \rightarrow \infty} |a_n| = \infty$, then is there an entire function f satisfying $A = Z_f$? The answer is yes.

Example. Let $A = \{a_1, \dots, a_n\}$ be a finite zero set. Then $f(z) = (z - a_1) \dots (z - a_n)$ and $A = Z_f$.

Proposition. If there are two entire functions f and g such that $Z_f = Z_g$, then $\frac{f}{g}$ is entire with no zeros. Since $\mathbf{C}$ is simply connected, there exists an entire function h such that $\frac{f}{g} = e^h$. That is, $f = ge^h$.

Example. An entire function f such that $Z_f = \mathbf{Z}$.

$$f(z) = \sin \pi z$$

How do we find an entire function f such that $Z_f = \{2, 3, 4, 5, 6, \dots\}$? Let's use the notation

$$\prod_{n=1}^N a_n = a_1 a_2 \dots a_N$$

and write

$$f(z) = \left(1 - \frac{z}{2}\right) \left(1 - \frac{z}{3}\right) \left(1 - \frac{z}{4}\right)$$

This function is zero at 1, and our conjecture is false. The idea is to find a multiplier $e^{g(z)}$ to take this property away!

$$f(z) = \prod_{n=2}^{\infty} \left(1 - \frac{z}{n}\right) e^{g(z)}$$

Lemma. If $a_n \geq 0$, then

$$\prod_{n=1}^{\infty} (1 + a_n)$$

converges if and only if

$$\sum_{n=1}^{\infty} a_n$$

converges.

Proof. If $x \geq 0$,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \geq 1 + x$$

□

Proof.

$$\sum_{n=1}^N a_n \leq \prod_{n=1}^N (1 + a_n) \leq \prod_{n=1}^N e^{a_n} = e^{\sum_{n=1}^N a_n}$$

using the comparison test. $\square$

Theorem. (Big theorem I) Let $a_n \in \mathbf{C}$. If $\sum_{n=1}^{\infty} |a_n| < \infty$, then $\prod_{n=1}^{\infty} (1 + a_n)$ converges and if $\prod_{n=1}^{\infty} (1 + a_n) = 0$ then $a_n = -1$ for some $n \in \mathbf{N}$.

Proof. Since $\sum_{n=1}^{\infty} |a_n|$ converges, there is $K \in \mathbf{N}$ such that if $n \geq K$, then $|a_n| < \frac{1}{2}$. Then

$$\prod_{n=k}^N (1 + a_n) = \prod_{n=k}^N e^{\log(1+a_n)} = e^{\sum_{n=k}^N \log(1+a_n)}$$

Notice that

$$\sum_{n=k}^N |\log(1 + a_n)| \leq \sum_{n=k}^N 2|a_n|$$

by lemma. Thus,

$$\sum_{n=k}^{\infty} \log(1 + a_n)$$

converges to some $A \in \mathbf{C}$.

$$\prod_{n=k}^N (1 + a_n) = e^A \neq 0$$

$\square$

Lemma. We denote $\log z$ for $z \neq 0$ and $-\pi < \arg z < \pi$.

$$\log z = \ln |z| + i \arg z$$

If $|z| < \frac{1}{2}$, then $\log(1 + z)$ satisfies $|\log(1 + z)| \leq 2|z|$.

Proof. Since $\log(1 + z)$ is analytic in $|z| < 1$,

$$\log(1 + z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4}$$

for $|z| < 1$. Thus, if $|z| < \frac{1}{2}$,

$$|\log(1 + z)| = \left| z - \frac{z^2}{2} + \frac{z^3}{3} + \dots \right| \leq |z| \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) = 2|z|$$

$\square$

Theorem. (Big theorem II) Let $\{f_n\}$ be analytic in a domain D . If there is $\{M_n\}$ with $M_n \geq 0$ satisfying $\sum_{n=1}^{\infty} M_n < \infty$ and $|f_n(z) - 1| \leq M_n$ for all $z \in D$, then

$$f(z) = \prod_{n=1}^{\infty} f_n(z)$$

is analytic in D since $\prod_{n=1}^N f_n(z)$ converges uniformly on every compact subset.

Proof. There exists $n \geq k$ such that $|f_n(z) - 1| < \frac{1}{2}$ for all $z \in D$.

$$\prod_{n=k}^N f_n(z) = \prod_{n=k}^N e^{\log(f_n(z))} = e^{\sum \log(f_n(z))} \leq e^{\sum M_n}$$

as $|\log(f_n(z) - 1)| \leq M_n$. □

Lemma. If f_n converges uniformly and g is uniformly continuous,

$$g \circ f_n$$

converges uniformly.

Example. Prove that $\prod_{n=1}^{\infty} \left(1 - \frac{z}{n^2}\right)$ is an entire function with $Z_f = \{n^2 \mid n \in \mathbf{N}\}$.

Proof. Let $D = D(0, R)$. Take $f_n(z) = z/n^2$ and $M_n = R/n^2$ we can conclude that $f(z)$ is analytic in D . □

LECTURE 21 (NOVEMBER 27TH)

Recall. Let f_n be analytic in a domain D . Let

$$\sum_{n=1}^{\infty} |f_n(z) - 1| \leq M_n$$

for all $z \in D$ where $\sum_{n=1}^{\infty} M_n < \infty$. Then,

$$f(z) = \prod_{n=1}^{\infty} f_n(z)$$

is analytic in D , and $f(z_0) = 0$ if and only if $f_n(z_0) = 0$ for some n .

Proof. There is a K such that $M_n < \frac{1}{2}$ for all $n \geq K$. Thus, if $n \geq K$, then $|f_n(z) - 1| < \frac{1}{2}$ for all $z \in D$ ($\frac{1}{2} \leq |f_n(z)| < \frac{3}{2}$ for all $z \in D$ if $n \geq K$). Thus, if $n \geq K$, then $\log f_n$ is analytic in D .

$$\prod_{n=k}^N f_n(z) = \prod_{n=k}^N e^{\log f_n(z)} = e^{\sum_{n=k}^N \log f_n(z)} \rightarrow e^{F(z)}$$

as $|\log f_n(z)| = |1 + \log f_n(z) - 1| \leq 2|f_n(z) - 1| \leq 2M_n$. If K is a compact subset of D , then $F(K)$ is compact and $g(z) = e^z$ is uniformly continuous on $F(K)$, which tells us that e^F is analytic in D . $\square$

Example. The function

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n^2}\right)$$

is an entire function with $Z_f = \{n^2 \mid n \in \mathbf{N}\}$. Let $D = D(0, R)$ and $f_n(z) = 1 - z/n^2$. Then,

$$\sum_{n=1}^{\infty} |1 - f_n(z)| \leq \sum_{n=1}^{\infty} \frac{R}{n^2}$$

setting $R/n^2 = M_n$, f is analytic on $D(0, R)$. Since R is arbitrary, f is entire.

Example. Let $0 \leq a_n < 1$. Then if $\sum_{n=1}^{\infty} a_n < \infty$, then $\prod_{n=1}^{\infty} (1 - a_n) > 0$. Put $f_n(z) = 1 - a_n$ constant.

Proposition. Let

$$A = \{1, 2, 3, 4, \dots\}$$

$$B = \{1, 4, 9, 25, \dots\}$$

$$C = \{1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \dots\}$$

$$D = \{\ln 1, \ln 2, \ln 3, \ln 4, \dots\}$$

we can find entire functions f_1, f_2, f_3, f_4 such that $Z_{f_1} = A, Z_{f_2} = B, Z_{f_3} = C, Z_{f_4} = D$.

Definition. The Weierstrass elementary factor $E_p(z)$ is defined by $E_0(z) = 1 - z$ for $p \in \mathbf{N}$ ($z \in \mathbf{C}$ and $p = 0, 1, \dots$),

$$E_p(z) = (1 - z) \exp\left(z + \frac{z^2}{2} + \dots + \frac{z^p}{p}\right)$$

If $z_n \neq 0$ and $|z_n| \rightarrow \infty$ then

$$\prod_{n=1}^{\infty} E_n\left(\frac{z}{z_n}\right)$$

is an entire function whose zero set is z_n (at worst, this is an entire function). Notice how $E_p(1) = 0$ and $E_p(0) = 1$.

Lemma. (Weierstrass) If $p \in \mathbf{N}$ and $|z| \leq 1$, then $|E_p(z) - 1| \leq |z|^{p+1}$.

Proof. Since E_p is entire,

$$E_p(z) = 1 + \sum_{n=1}^{\infty} a_n z^n$$

for all z . Thus,

$$E_p'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

By straightforward calculation,

$$\begin{aligned} E_p'(z) &= -\exp\left(z + \frac{z^p}{p}\right) + (1-z) \exp\left(z + \dots + \frac{z^p}{p}\right) \times (1+z+z^2+\dots+z^{p-1}) \\ &= -\exp\left(z + \dots + \frac{z^p}{p}\right) + (1-z^p) \exp\left(z + \dots + \frac{z^p}{p}\right) \\ &= -z^p \exp\left(z + \frac{z^2}{2} + \dots + \frac{z^p}{p}\right) \end{aligned}$$

From this we know that

$$a_1 = a_2 = \dots = a_p = 0$$

and $a_n \leq 0$ for all $n \geq p+1$. Hence, for $|z| \leq 1$,

$$\begin{aligned} |E_p(z) - 1| &= \left| \sum_{n=1}^{\infty} a_n z^n \right| \\ &= \left| \sum_{n=p+1}^{\infty} a_n z^n \right| \\ &= |z|^{p+1} \left| \sum_{n=p+1}^{\infty} a_n z^{n-p-1} \right| \\ &\leq |z|^{p+1} \cdot \sum_{n=p+1}^{\infty} |a_n| \end{aligned}$$

Since $E_p(1) = 0$, we have $1 + \sum_{n=1}^{\infty} a_n = 0$ and thus

$$1 + \sum_{n=p+1}^{\infty} a_n = 0$$

since $a_1 = \dots = a_p = 0$. Thus

$$1 - \sum_{n=p+1}^{\infty} |a_n| = 0$$

since $|a_n| = -a_n$ for $n \geq p+1$. Then,

$$\sum_{n=p+1}^{\infty} |a_n| = 1$$

and

$$|E_p(z) - 1| \leq |z|^{p+1}$$

□

Theorem. (Big theorem) Let $\{z_n\}$ be a sequence in $\mathbf{C}$ such that $z_n \neq 0$ and $|z_n| \rightarrow \infty$. If

$$\sum_{n=1}^{\infty} \frac{1}{|z_n|^{p+1}} < \infty$$

for $p = 0, 1, 2, \dots$ then

$$f(z) = \prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is an entire function with

$$Z_f = \{z_n \mid n \in \mathbf{N}\}$$

Also,

$$\prod_{n=1}^{\infty} E_n\left(\frac{z}{z_n}\right)$$

is an entire function whose zero set is $\{z_n\}$.

Proof. For every $z \in \bar{D}(0, R)$,

$$\left|E_p\left(\frac{z}{z_n}\right) - 1\right| \leq \left|\frac{z}{z_n}\right|^{p+1} \leq 1$$

for $|z/z_n| \leq 1$. Notice that this is satisfied for all $n \geq K$ since $|z_n| \rightarrow \infty$. And,

$$\sum_{n=k}^{\infty} \left|\frac{z}{z_n}\right|^{p+1} = |z|^{p+1} \sum_{n=k}^{\infty} \frac{1}{|z_n|^{p+1}} \leq M \sum_{n=k}^{\infty} \frac{1}{|z_n|^{p+1}}$$

Thus, by the previous theorem,

$$\prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is analytic in $D(0, R)$ for every R . Hence,

$$\prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is entire whose zero set is $\{z_n\}$.

□

Example. Using this theorem, show that an entire function f such that $Z_f = \{1, 2, \dots\}$ is

$$f(z) = \prod_{n=1}^{\infty} E_1\left(\frac{z}{z_n}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n}\right) e^{z/n}$$

An entire function g such that $Z_g = \{1, 4, 9, 16, 25, \dots\}$ is

$$g(z) = \prod_{n=1}^{\infty} E_0\left(\frac{z}{n^2}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n^2}\right)$$

Similarly $Z_h = \{1, \sqrt{2}, \sqrt{3}, \dots\}$.

$$h(z) = \prod_{n=1}^{\infty} E_2\left(\frac{z}{\sqrt{n}}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{\sqrt{n}}\right) e^{z/\sqrt{n} + z^2/2n}$$

If $Z_F = \{\ln 1, \ln 2, \ln 3, \dots\}$, there exists no such p , and we must say that

$$F = \prod_{n=2}^{\infty} E_n\left(\frac{z}{\ln n}\right) = z \prod_{n=2}^{\infty} \left(1 - \frac{1}{\ln n}\right) \exp\left(\frac{z}{\ln n} + \frac{1}{2}\left(\frac{z}{\ln n}\right)^2 + \dots + \frac{1}{n}\left(\frac{z}{\ln n}\right)^n\right)$$